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Research Statement

My research lies at the intersection of quantum Shannon theory and entanglement theory, focusing on how symmetries and noise shape the fundamental limits of quantum communication and storage. A central contribution is a proof of the PPT² conjecture for a broad class of quantum channels with diagonal unitary symmetry, clarifying when entanglement swapping cannot preserve entanglement for quantum repeater protocols [1]. This result forms part of my broader investigation of diagonal symmetries in quantum theory, where I have systematically studied the structure and properties of diagonally invariant states and channels [2].

In quantum Shannon theory, I have developed an elementary perturbative approach to detecting positive quantum capacity—the ability of a noisy channel to transmit qubits reliably at a nonzero rate. This approach yields practical tests based on comparisons between input, output and environment dimensions, as well as structural characterizations of certain zero-capacity channels [3]. More recently, I have studied information storage and transmission under Markovian noise, where noise accumulates over time in a quantum memory or along a communication channel. My work provides explicit, efficiently computable capacity bounds and establishes strong additivity and strong converse results in the limit of long storage times or transmission distances [4]. Related work addresses the discrimination of different Markovian noise processes in this same limit [5].

My current research focuses on identification via quantum channels, where the receiver tests whether a transmitted message matches a specified identity. I use tools from high-dimensional probability and geometry to investigate the communication capabilities of this task and establish fundamental limits on its performance [6, 7]. Alongside this, I study how hybrid classical–quantum memories can be encoded, transmitted and recovered through noisy channels [8]. My broader goal is to develop a theory of task-dependent protection that tailors encoding and recovery to the information needed for a prescribed task, achieving the required accuracy with fewer physical resources.

References

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